# Research Direction Memo

When to Correct? Statistical Timing of Corrective Refinement

Supervisor-facing draft

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## Proposed thesis question

The thesis studies correction timing in masked diffusion as a finite-budget statistical decision problem. The predictor schedule, corrector mechanism, and quality metric are fixed. The object of study is when a limited number of corrective refinement steps should be spent along the generation trajectory.

The main research question is:

> When should a masked diffusion sampler spend a limited correction budget, and can correction timing be predicted from the current generation state?

A more formal version is:

> For masked diffusion language models with a finite number of corrective refinement steps, can the time steps at which correction is most valuable be characterized and predicted using observable uncertainty and error signals from the generation trajectory?

The interaction question remains part of the thesis, but as an explanatory sub-question rather than the headline:

> Are correction gains approximately additive over time, or do redundancy and interactions between correction steps explain why simple timing rules fail?

## Why this is the right framing

This framing keeps the thesis close to statistics. The core quantities are potential outcomes, conditional expectations, held-out prediction error, finite-budget regret certificates, and interaction diagnostics. Search algorithms are useful, but they are not the main contribution. They act as empirical probes showing whether useful correction-timing structure exists beyond simple uncertainty rankers.

It also separates three questions that should not be conflated:

1. **Correction value:** where in generation does correction help?
2. **State predictability:** can observable trajectory statistics predict that value beyond absolute time?
3. **Interaction structure:** if simple marginal rules fail, is the failure caused by redundancy, higher-order interactions, or lack of stable local structure?

This is different from training a new corrector or studying which tokens to revise. PRISM, locally balanced proposals, and informed-corrector kernels are important related work for how to design informed correction moves. Here the correction move is fixed; the thesis studies allocation of that move across time.

## Statistical setup

Let $\mathcal{T} = \{1, \ldots, T\}$. A correction schedule is a subset $S \subseteq \mathcal{T}$, with $|S| = B$. Fix a masked-diffusion predictor, a corrector kernel, a coupling variable $\omega$ for common random numbers, and an integrable terminal utility $F : \mathcal{Y} \to \mathbb{R}$. The final sample under schedule $S$ is

$$Y^S = \Phi(S, \omega),$$

and the population schedule value is

$$J(S) = \mathbb{E}[F(Y^S)], \qquad G(S) = J(S) - J(\emptyset).$$

For paired experiments, seed-level potential outcomes are

$$G_i(S) = F(Y_i^S) - F(Y_i^\emptyset),$$

where the same seed $i$ is reused across schedules. This makes correction timing a potential-outcome problem: the counterfactual is what the same generation trajectory would have produced with a correction inserted at a different time.

**Marginal correction value.** For a single correction at time $t$,

$$\Delta_t = J(\{t\}) - J(\emptyset), \qquad \delta_{i,t} = G_i(\{t\}).$$

$\Delta_t$ answers the descriptive question: at which generation stages is a single correction most valuable on average?

**State-predictive correction value.** Let $Z_{i,t}$ be observable state features before deciding whether to correct at time $t$: diffusion phase, mask ratio, entropy, top-1/top-2 margin, recently unmasked fraction, token instability, likelihood change, or pre/post prediction disagreement when available. The statistical target is

$$m(t, z) = \mathbb{E}[\delta_t \mid t, Z_t = z].$$

The key question is whether $Z_t$ improves prediction of $\delta_t$ beyond time $t$ alone.

**Budgeted timing rule.** Given a predictor $\widehat{m}(t, Z_t)$, the simplest offline timing rule selects the top $B$ time steps by predicted correction value. An online trigger is the sequential version: spend correction budget only when $\widehat{m}(t, Z_t)$ is high enough relative to the remaining time and remaining budget.

## Theorem candidates

**Proposition 1** (State information as prediction gain)**.** *Let $\delta \in L^2$ be the marginal correction value for a randomly sampled trajectory-time pair. Let $\mathcal{F}_0 = \sigma(t)$ be the information in time alone and let $\mathcal{F}_1 = \sigma(t, Z_t)$, with $\mathcal{F}_0 \subseteq \mathcal{F}_1$. The Bayes predictors under squared loss are*

$$m_0(t) = \mathbb{E}[\delta \mid \mathcal{F}_0], \qquad m_1(t, Z_t) = \mathbb{E}[\delta \mid \mathcal{F}_1].$$

*Their mean-squared prediction errors satisfy*

$$\mathbb{E}[(\delta - m_0)^2] - \mathbb{E}[(\delta - m_1)^2] = \mathbb{E}[(m_1 - m_0)^2] \geq 0.$$

*Proof sketch.* Conditional expectation is the $L^2$ projection onto the corresponding information set. Since $\mathcal{F}_0 \subseteq \mathcal{F}_1$, the Pythagorean identity for nested projections gives the displayed equality. □

**Use in the thesis.** This gives the clean statistical meaning of the state-predictability question. If held-out prediction using $Z_t$ does not beat time-only prediction, then a state-adaptive trigger is not justified. If it does, the improvement measures how much information the current generation state carries about correction value.

**Proposition 2** (Finite-pool regret certificate for predicted marginal timing)**.** *Let $\mathcal{C}_B \subseteq \{S \subseteq \mathcal{T} : |S| = B\}$ be a fixed candidate pool and let*

$$A(S) = \sum_{t \in S} \Delta_t, \qquad \widehat{A}(S) = \sum_{t \in S} \widehat{m}(t, Z_t).$$

*Assume*

$$\sup_{S \in \mathcal{C}_B} |G(S) - A(S)| \leq \eta, \qquad \sup_{S \in \mathcal{C}_B} |A(S) - \widehat{A}(S)| \leq \alpha.$$

*If $S_G^\star \in \arg\max_{S \in \mathcal{C}_B} G(S)$ and $\widehat{S} \in \arg\max_{S \in \mathcal{C}_B} \widehat{A}(S)$, then*

$$G(S_G^\star) - G(\widehat{S}) \leq 2\eta + 2\alpha.$$

*Proof sketch.* Apply the generic finite-pool surrogate argument with surrogate $A$ and estimated surrogate $\widehat{A}$. The factor 2 appears because the surrogate error is paid once at the oracle schedule and once at the selected schedule. □

**Use in the thesis.** This theorem clarifies when a lightweight timing rule can be defended: correction gains must be approximately additive on the candidate schedules and the state predictor must be accurate enough on held-out trajectories. If either condition fails, the failure is a statistical result, not an implementation failure.

**Lemma 1** (Interaction diagnostic)**.** *For $s \neq t$, define the pair interaction*

$$\xi_{s,t} = G(\{s, t\}) - \Delta_s - \Delta_t.$$

*For a quadratic approximation*

$$Q_2(S) = \sum_{t \in S} a_t + \sum_{\{s,t\} \subseteq S} b_{s,t},$$

*the diminishing-return gap for $A \subseteq B \subseteq \mathcal{T}$ and $x \notin B$ is*

$$\big[Q_2(A \cup \{x\}) - Q_2(A)\big] - \big[Q_2(B \cup \{x\}) - Q_2(B)\big] = -\sum_{t \in B \setminus A} b_{x,t}.$$

*Thus negative pair coefficients imply diminishing returns for the quadratic approximation, but not automatically for the true $G$.*

*Proof sketch.* The marginal gain of adding $x$ to $A$ is $a_x + \sum_{t \in A} b_{x,t}$. Subtract the corresponding expression for $B$. A claim about $G$ needs residual control between $G$ and $Q_2$. $\square$

**Use in the thesis.** This is the formal bridge from state-predicted marginal timing to interaction diagnostics. If marginal or state-adaptive timing fails, $\xi_{s,t}$, additivity residuals, and one-swap neighborhoods explain whether the failure is due to redundancy, higher-order effects, or an essentially black-box landscape.

## Empirical plan

**RQ1: Where is correction valuable?** Estimate $\Delta_t$ over generation time using paired common-random-number runs. Report curves, confidence intervals, and phase summaries.

**RQ2: Is correction value predictable from state?** Compare time-only prediction $m_0(t)$ against state-based prediction $m_1(t, Z_t)$. Features can include mask ratio, entropy, margin, recently unmasked fraction, token instability, likelihood change, and prediction disagreement. Use held-out trajectories; do not evaluate a trigger on the same data used to choose it.

**RQ3: Can simple timing rules spend budget well?** Compare uniform, early-heavy, late-heavy, marginal-greedy, state-triggered, and oracle schedules under the same budget $B$. A learned logistic or small MLP trigger is optional and only justified if RQ2 shows held-out predictive signal.

**RQ4: Why do simple rules fail?** Measure additivity residuals, pair interactions, diminishing-return gaps, and one-swap neighborhood improvements. This is the interaction/redundancy part of the thesis.

## What current evidence already answers

Many of the empirical questions are already partly answered on ProSeCo-OWT. The thesis does not need to pretend these are open; the supervisor-facing story can present the current evidence as the beginning of the result.

| Question | Current answer on ProSeCo-OWT | Status |
|---|---|---|
| Is correction timing non-vacuous? | Yes. Existing paired experiments show MC-oracle headroom over uniform of about +0.45 at $B \in \{2, 3, 4\}$, with a K=30 replication around +0.37. | Answered for ProSeCo-OWT. |
| Do simple uncertainty/ranker signals recover the headroom? | No. Tested separable rankers fail to recover the MC-oracle headroom. The mean marginal envelope becomes weak by larger budgets. | Answered negatively for tested signals. |
| Can simple adaptive state conditioning solve it? | Current evidence says no. The bucketed adaptive-controller pilot shrinks calibration error by less than 1.7% and does not recover headroom. The held-out state-predictability audit also finds no improvement from adding current state features to time-only prediction. | Answered negatively for current features. |
| Is there useful structure beyond simple rankers? | Yes. True-$G$ search procedures recover much of the oracle headroom: CD-G about 74–84%, BS-AG about 49–64% of the canonical headroom at $B \in \{2, 3, 4\}$. | Answered positively. |
| Are interactions present? | Yes. Pair interactions are structured and mostly negative / redundant. Cheap pairwise composition helps at $B = 2$ but fails at $B = 3, 4$. | Answered as diagnostic result. |
| Is the landscape smooth enough for BO or submodular theory? | Not clearly. One-swap smoothness exists but is weak, diminishing-return gaps are mildly positive but region-dependent, and BO remains unclear. | Answered conservatively. |

This means the current thesis is not a speculative proposal from zero. It is closer to a synthesis and formalization of an empirical story already visible in the results:

> Correction timing matters on ProSeCo-OWT. Simple observable uncertainty signals do not reliably predict the best timing. The failure is not random: there is structured redundancy and local search structure, but not enough evidence for a clean submodular or Bayesian-optimization theory.

## Model choice and external validity

| Model / codebase | Role | Reason |
|---|---|---|
| ProSeCo-OWT | Main case study | Best fit. It is a self-correcting masked-diffusion backend with an available small OpenWebText checkpoint and a fixed correction mechanism. It lets the thesis study timing without training a corrector from scratch. |
| MDLM-OWT | Baseline only | Feasible and instrumentable, but only relevant if a meaningful correction intervention is defined. Without that, it is a predictor model rather than a self-correction timing platform. |
| Informed-correctors code | Controlled fallback | Theoretically aligned with predictor-corrector sampling and informed correctors. Less aligned with language-scale ProSeCo, so best as a small controlled appendix if needed. |
| SEDD | Optional comparison | Important discrete diffusion model, but not naturally a ProSeCo-style self-correcting masked sampler. |
| ProSeCo-LLaDA / LLaDA 8B | Stretch only | Scientifically relevant but infrastructure-heavy. Existing LLaDA evidence in this project did not show clear positive oracle headroom, so it should not be the core model. |

**Does it make sense to try the heavier model?** Only as an external-validity gate, not as the main path. The heavier LLaDA/ProSeCo-LLaDA direction is scientifically attractive, but it has high infrastructure risk and current local evidence is inconclusive: the bounded LLaDA-SFT probe did not transfer the positive MC-oracle headroom at tested resolution. Before running a full heavy-model scheduling study, the necessary gate is a small pilot showing:

- positive correction headroom over uniform with a meaningful confidence interval;
- non-degenerate corrector behavior;
- a stable and interpretable utility metric.

Without this gate, a heavier model is likely to consume time without improving the thesis argument.

## What is model-agnostic, and what is not

The theory is model-agnostic in a precise but limited sense. The definitions of $J(S)$, $G(S)$, $\Delta_t$, $m(t,z)$, additivity error, pair interactions, and finite-pool regret apply to any fixed tuple:

$$(\text{predictor, corrector, utility } F\text{, budget } B).$$

They do not require ProSeCo, reversibility, detailed balance, or a specific neural architecture.

The empirical classification is not model-agnostic. The statement "simple signals fail and search recovers headroom" is currently a ProSeCo-OWT result. A different model/corrector/metric could lie in a different regime:

- no-op: correction has no measurable headroom;

- marginal: time or uncertainty signals predict correction value well;
- interaction/redundancy: marginal signals are insufficient but interactions are structured;
- higher-order/search: only direct true-$G$ search finds the good schedules.

Thus the model-agnostic output of the thesis is the diagnostic framework and the finite-pool certificates. The ProSeCo-OWT output is a case-study classification within that framework.

## Recommended next steps

Given the work already completed, the evidence does not support starting a major new online scheduler project. The simple online/state conditioning and held-out state-predictability tests are negative, and the landscape evidence says Bayesian optimization is still unclear. The most defensible next steps are:

1. **Write the thesis story around the results already obtained.** The current results already answer the main diagnostic questions on ProSeCo-OWT. The immediate task is synthesis, not another large experiment.
2. **Do not build an online scheduler from the current feature set.** An online scheduler is justified only if current-state features add held-out predictive information beyond time. The completed audit does not show this.
3. **Use the state-predictability negative as a result.** The negative audit is useful evidence: it shows that the current observable uncertainty/error features do not explain correction timing enough to support a simple adaptive trigger.
4. **Use heavier models only for a feasibility gate.** Run LLaDA/ProSeCo-LLaDA only if external validity becomes mandatory and only after a small headroom pilot passes.
5. **Prepare the supervisor discussion around scope.** The key decision is whether a ProSeCo-OWT case study plus model-agnostic diagnostic framework is accepted as the thesis contribution, or whether the supervisor requires one additional external-validity check.

## Source anchors


- Zhao, Shi, Chen, Druckmann, Mackey, and Linderman, *Informed Correctors for Discrete Diffusion Models*, arXiv:2407.21243.
- Zanella, *Informed proposals for local MCMC in discrete spaces*, arXiv:1711.07424 / JASA 2020.
- Lavenant and Zanella, *Error Bounds and Optimal Schedules for Masked Diffusions with Factorized Approximations*, arXiv:2510.25544.
- Kim et al., *Fine-Tuning Masked Diffusion for Provable Self-Correction*, arXiv:2510.01384.
- Schiff et al.,
*Learn from Your Mistakes: Self-Correcting Masked Diffusion Models*,
arXiv:2602.11590.